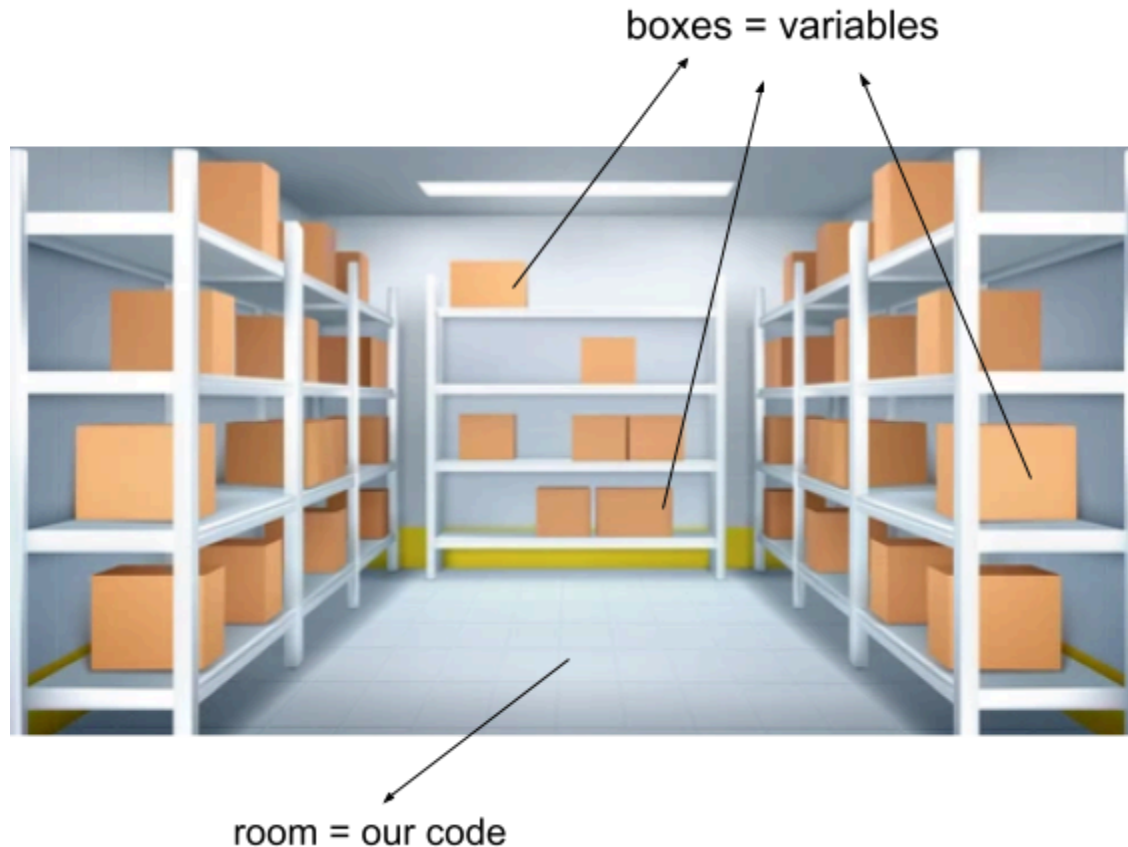


Variable - a way for us to store and share important information

Understanding variables...



Let's say we have a storage room which represents **our code** or where we are writing our code. The different boxes on the shelves each have labels on them. The label is used so we remember what's in the box. We can put something in the box and whenever we need the item in that box we can take it out by seeing the label.

The boxes on the shelves are **variables** in our code. The labels we use on the boxes are the **names of the variables**. The items we put in our boxes are the **important pieces of information we store**.

That's why it is important to name our variables that are relevant to the context of our code. It will make it easier to understand when we need an item in the variable.

For example...

We made code for a video game (storage room) and needed a variable (box) to store the player's score (important information). The player's score right now is 5.

Which one would be easier to understand:

X = 5

OR

_playerScore = 5

Probably the second one right?

Naming Variables:

- The variable name must start with a lowercase letter or an underscore.
- The variable name cannot start with a number.
- Variable names cannot contain special characters. (Example: &%\$@)

Examples of **GOOD** variable names:

name = "Sunny"

_playerScore = 5

heightInCentimeters = 79.4

Examples of **BAD** variable names:

Name = "Sunny"

%playerScore = 5

3heightInCentimeters = 79.4

Important Concept - **Syntax**

You might be wondering up until this point how our computer understands the code we are writing or even why we need to structure our code the way we need to.

Like why do we need = signs when making a variable?

Well our computers can't understand human language like words (or else none of this would be needed). So to instruct our computer to do something, we write code which is a language that computers can understand.

But it's never that simple :(

Computers are really picky. So if you don't follow the rules or format the computer wants in your code, it won't understand what you are trying to instruct it to do and simply won't run or execute your code.

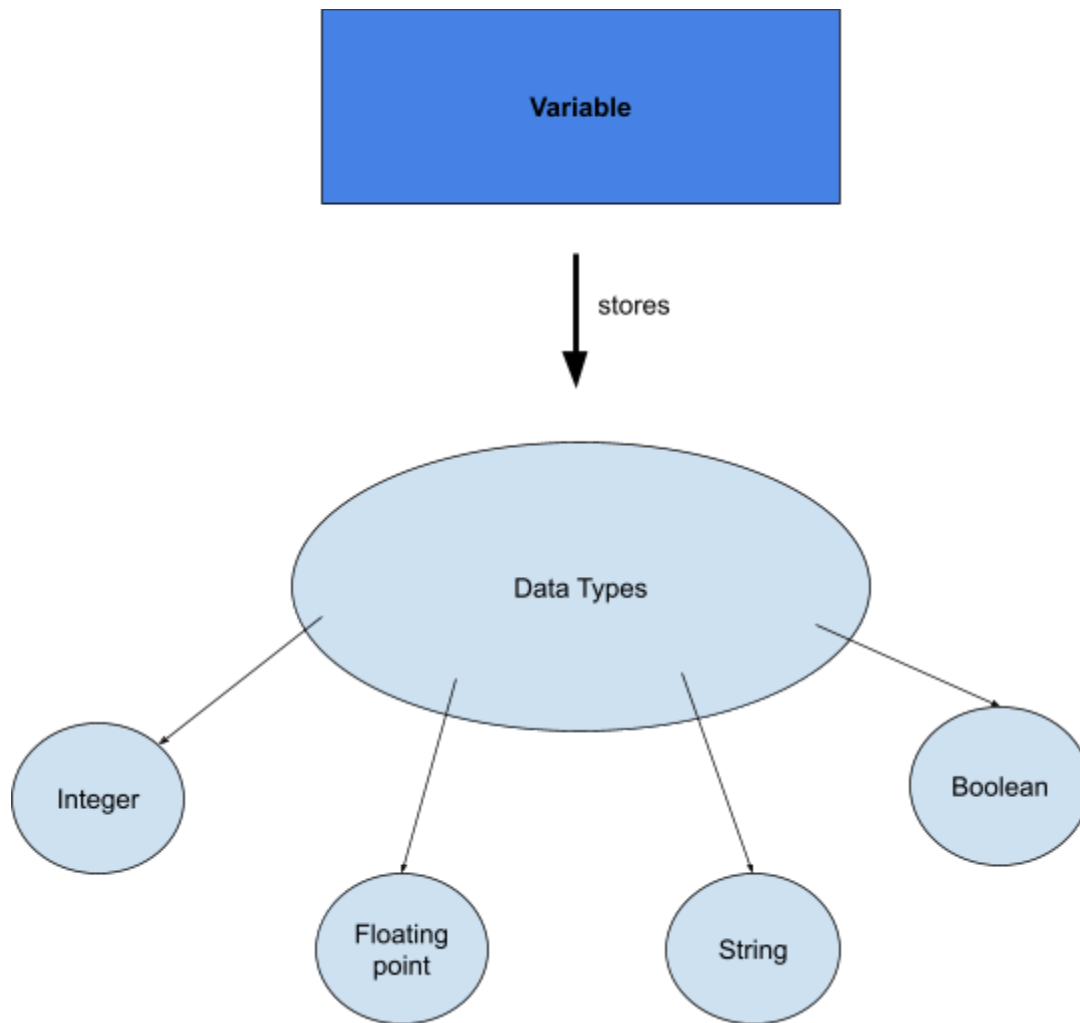
When running a block or piece of code you may see something in the output error, showing a syntax error. This is because you didn't follow the rules the computer was expecting when running that piece of code.

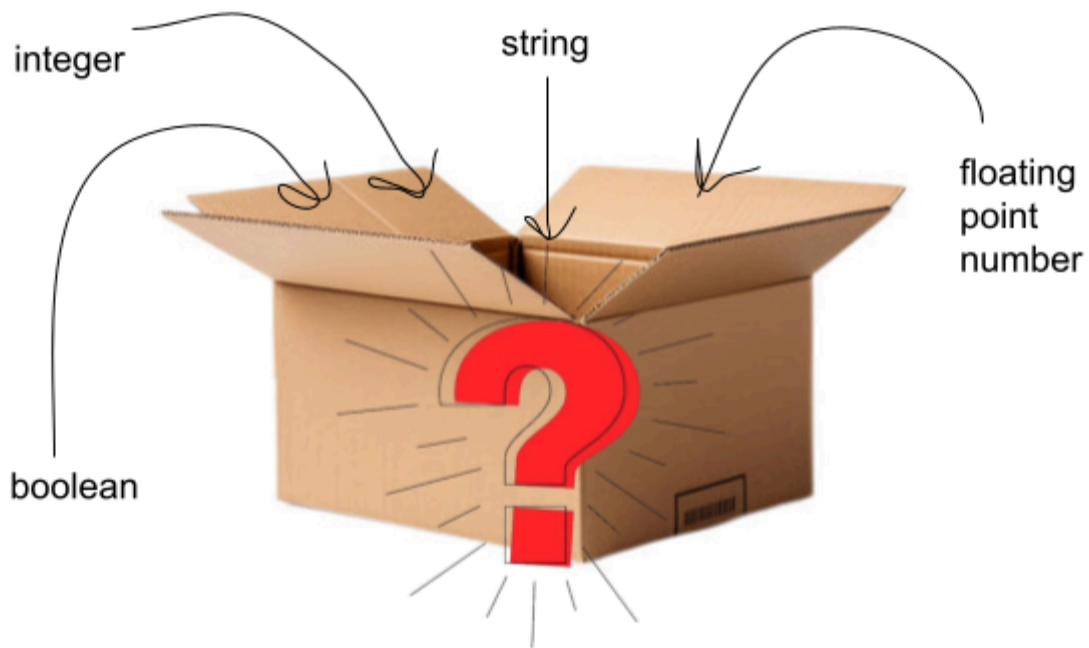
The computer cuts you some slack though, and most code editors (where you write code) will tell you where you messed up when formatting your code.

Now we know we can make a **variable** (box) and place it in **our code** (the storage room)

But why would the variable (box) matter if there's *nothing* in it?

We have to enter some type of **data** (important pieces of information) to a variable (box) for the variable to matter.





*Breaking down each **MAIN** Data Type...*

Integer - a WHOLE number that can either be negative or positive

ex. `playerScore = 5`

Float (floating point number) - a DECIMAL number that can either be negative or positive

ex. `heightInCentimeters = 98.2`

Boolean - used to represent a True or False value

ex. `15 < 20 → True`

String - a set of CHARACTERS surrounded with double or single quotes

ex. `name = "Sunny"`